Section 3: Regular Expressions (REGEX)

21 questions — using Pattern/Matcher, matches(), replaceAll(), split() with regex patterns

Q26 | 2020 (COVID - single packet)

Which of the following represents the output of the code segment shown on the right?

A) true true true
B) true true false
C) true false false
D) false true true
E) false false true

```
String s = "325-555-1234";
out.print(s.matches(".{10}") + " ");
out.print(s.matches("\\d+-\\d+-\\d+")+" ");
out.print(s.matches("325\\W555\\S1234"));
```

Q32 | 2020 (COVID - single packet)

What is the output of the code segment shown on the right?

A) de#yk
B) adcxdz
C) ac9e#3py7xz6yk
D) 3
E) adc9de# py7xdz6yk

```
Scanner s = new
```

Scanner("adc9de#3py7xdz6yk");

```
s.useDelimiter("\\d");
while(s.hasNext())
if(s.next().matches("\\w{3}"))
out.print(s.next());
```

Q28 | 2025 Invitational A

What is the output by the code to the right?

A) true
B) false
C) Output cannot be determined until runtime
D) There is no output due to a compile error
E) There is no output due to a runtime error

```
String s1 = "H3llo Th3r3!";
String s2 = "H..{2,4}\\S..{2,5}";
```

s1 = "" + s1.matches(s2);

```
out.println(s1);
```

Q28 | 2025 Invitational B

What is the output by the code to the right?

A) true true
B) false true
C) true false
D) false false

E) There is no output due to a runtime error

```
String s = "Luke Skywalker";
String fin = "";
String r = "(\\w)+|(\\s){0,3}";
fin += s.matches(r) + " ";
r = "([A-z]+ ?)*";
fin += s.matches(r);
out.println(fin);
```

Q20 | 2025 State

What is output by the code to the right?

A) true true
B) true false
C) false true
D) false false
E) There is no output due to a runtime error

```
String m1 = "(\\w+ ){2,7}";
String m2 = "([A-Z]?[a-z]* ?)+";
String s = "One of a Kind";
out.print(s.matches(m1));
out.print(" ");
out.print(s.matches(m2));
```

Q23 | 2024 Invitational B

What is the output of the code segment shown on the right?

A) 22
B) 23
C) 24
D) 25
E) 26

```
String Q = "ABCDEFGHIJKLM";
String R = "NOPQRSTUVWXYZ";
String T = R + Q;
for(int x=0; x<T.length(); x++)
{
String Z = T.substring(x,x+1);
if (Z.matches("[TEXAS]"))
T=T.substring(0,x)+T.substring(x+1);
}
out.print(T.length());
```

Q24 | 2024 State

What is the output of the code segment to the right?

A) RSITY
B) V
C) V
D) TY
E) RIT

```
ArrayList<String>Starsky;
Starsky = new ArrayList<String>();
```

```
String Hutch = "UNIVERSITY";
for(int x=0; x<Hutch.length(); x++)
Starsky.add(Hutch.substring(x));
for(int x=0; x<Starsky.size(); x++)
{
String A = Starsky.get(x);
if(A.matches("[AEIOU].*"))
Starsky.remove(x);
}
out.print(Starsky.get(2));
```

Q37 | 2023 Region

What is the output of the code segment shown on the right?

A) 1
B) 2
C) 3
D) 4
E) 5

```
int C=0;
String A = "R2-D2";
if(A.matches(".....")) C++;
if(A.matches(".2.2")) C++;
if(A.matches("[A-T]*")) C++;
if(A.matches("2.*")) C++;
if(A.matches(".*2")) C++;
if(A.matches(".*[0-9].*[0-9]")) C++;
out.print(C);
```

Q23 | 2023 State

What is the output of the code segment shown on the right?

A) 4
B) 6
C) 8
D) 10
E) 12

```
String St = "BEAR OWL DOG CAT LION ";
```

St += "ZEBRA RAT PIG TIGER GORILLA";

```
int N = 0;
Scanner Sue = new Scanner(St);
while(Sue.hasNext())
{
Sue.next();
String A = Sue.next();
if(A.matches("..O.")) N++;
if(A.matches("[A-C].*")) N++;
if(A.matches("..G")) N++;
if(A.matches(".*R.*")) N++;
}
out.print(N);
```

Q39 | 2022 Region

How many unique strings could replace *CODE*\ for the code below to output true? Write your answer in the blank provided

on the answer document.

```
String A = "[ABC]{1,2}";
String B = "[BCD]{2,3}";
String C = \*CODE*\;
out.print(C.matches(A) && C.matches(B));
```

Q40 | 2022 State

What is a String that could replace *CODE*\ for the code below to output true? Write your answer in the blank provided on the
answer document.

```
String A = "A{1,5}B{1,5}C{1,5}";
String B = ".*ABC*";
String C = "....B...";
String S = /*CODE*/;
boolean x = S.matches(A);
boolean y = S.matches(B);
boolean z = S.matches(C);
out.println(x && y && z);
```

KEY

Q36 | 2021 Invitational A

What is the output of the code segment shown on the right?

A) 12
B) 5
C) 6
D) 10
E) 7

```
String str = "UIL_Comp^Sci-2021";
int r = 0;
for(int i = 0; i < str.length(); i++)
if(str.substring(i, i + 1).matches("[^a-z]"))
```

r = r + 1;

```
out.print(r);
```

Q18 | 2021 Region

What is the output of the code segment shown on the right?

A) [abc, zm, o]
B) [abc, tzm, no]
C) [abcxytzmxyno]
D) [xyt, xyn]
E) [abcxy, zmxy, o]

```
String r = "xy.";
String s = "abcxytzmxyno";
Pattern p = Pattern.compile(r);
String i[] = p.split(s);
out.println(Arrays.toString(i));
```

Q16 | 2019 Invitational B

How many times does the code segment shown on the right
print true?

A) 0
B) 15
C) 3
D) 4
E) 18

```
String str="abc^245#s&890jhy%165x";
for(int i=0;i<str.length()-3;i++) {
String s=str.substring(i, i+3);
out.println(s.matches("[a-z]\\W\\d"));
}
```

Q17 | 2019 District

What is the output of the code segment shown here?

```
String str="325-978-1400";
out.print(str.matches(".{3}-\\w+-\\S+")+" ");
out.print(str.matches("325.978.1400")+" ");
out.print(str.matches(".*"));
```

A) true true true
B) true false true
C) true true false
D) false true false
E) false false false

Q19 | 2019 State

Which line in the code segment shown here will print true?

```
final int flags=Pattern.CASE_INSENSITIVE|Pattern.LITERAL;
Pattern p=Pattern.compile("[abc]+", flags);
out.println(p.matcher("[aBc]+").matches());//line #1
out.println(p.matcher("abc").matches());//line #2
out.println(p.matcher("aBcAbC").matches());//line #3
```

A) line #1
B) line #2
C) line #3
D)
E) None of the above

Q16 | 2018 Invitational A

What is the output of the code segment shown here?

```
String s="abcde";
for(int i=0;i<s.length();i++)
out.print(s.substring(i, i+1).matches("[^aeiou]")+" ");
}
```

A) true false false false true
B) false true true true false
C) false false false false false
D) true false false false false
E) false true true true true

Q22 | 2018 District

What is the output of the code segment shown on the right?

A) true true false

B) true false true
C) false true true
D) false false true
E) false true false

```
out.print("123ABC".matches("\\D{3}\\W{3}")+" ");
out.print("555-5555".matches(".+")+" ");
out.print("Alphabet".matches("A[a-z]?"));
```

Q35 | 2018 Region

Which of the following best describes the value returned by method?

A) The number of vowels in the string s or -1 if s does not contain a vowel.
B) The index number of the first vowel contained in s or -1 if s does not contain a vowel.
C) The index value of each vowel in s or -1 if there are no vowels present.
D) The ordinal (1 based) of the first vowel in in s or -1 if s does not contain a vowel.
E) The number of times the string “aeiou” is found in s or -1 if it does not occur.

```
   public static int method(String s) {
s=s.toLowerCase();
   int x;
   for(x=0;x<s.length();x++)
   if(s.substring(x, x+1).matches("[aeiou]"))
break;
   if(x!=s.length())
   return x+1;
   else
   return -1;
   }
```

Q20 | 2018 State

What is the output of the code segment shown here?

```
   String s="uil.academics@uiltexas.org";
   out.print(s.matches(".+"));
   out.print(s.matches("\\S{3}.\\D+@[a-z]+.\\w{3}"));
   out.print(s.matches("uil\\.?\\S*@+uiltexas.org"));
```

A) falsefalsetrue
B) falsetruetrue
C) truefalsetrue
D) truetruetrue
E) falsefalsefalse

Regular Expressions

2020 (COVID - single packet)
Q26: D Q32: A

2025 Invitational A
Q28: A

2025 Invitational B
Q28: B

2025 State
Q20: C

2024 Invitational B
Q23: A

2024 State
Q24: A

2023 Region
Q37: C

2023 State
Q23: A

2022 Region
Q39: 4

2022 State
Q40: AAAABCCC

2021 Invitational A
Q36: 12

2021 Region
Q18: A

2019 Invitational B
Q16: 3

2019 District
Q17: A

2019 State
Q19: A

2018 Invitational A
Q16: B

2018 District

Q22: E

2018 Region

Q35: D

2018 State

Q20: D